

Planning a Bank-Hosted AI Deployment

How to turn an intended banking workflow into a scoped operating arrangement, with tested capacity and a workable recovery path

Defined work. Tested capacity. Clear ownership.

A bank-hosted deployment needs more than a model and a server. This guide helps technology, business, and vendor-management teams define the workload, test the proposed environment, and decide what may proceed before a proposal is finalized.

01

Scope the workload

02

Test the environment

03

Plan support and recovery

For CIOs, technology leaders, project sponsors, and vendor-management teams

An educational planning guide with a worked deployment example and a concise decision brief.

01 / START WITH THE OPERATING NEED

A deployment is an arrangement, not a model.

The useful planning question is not which model sounds most capable. It is what work the bank wants to support, under which conditions, and with whose responsibility.

A bank-hosted arrangement can keep selected processing within an environment the bank controls, but it still has identities, source systems, updates, support paths, and operating dependencies. “Hosted at the bank” does not establish that every optional function stays local or that every feature in a general desktop product is included.

Planning question	What the answer should make visible
What work will it support?	The task, users, expected volume, source types, accepted deliverable, and actions deliberately excluded.
What must it dependably do?	Quality criteria, response or completion expectations, review requirements, and consequences of delay or failure.
Where will it operate?	The proposed environment, connected systems, optional external paths, and relevant support dependencies.
Who owns the arrangement?	Business, technology, security, operations, support, and vendor-management responsibilities.
How will the bank decide?	Evidence required for evaluation, limited production, expansion, or rejection, with named decision authority.

Separate the evaluation from the target state

A synthetic evaluation, a controlled pilot, and an operational service can use the same software while having very different volume, information, permissions, and recovery needs. Document each stage separately. Do not use a successful small demonstration to infer capacity, resilience, or readiness for a broader workload.

Plan the complete operating path.

Request → approved source access → processing → draft result → human review → return or action → records and recovery. A deployment decision should explain who owns each step and what happens when it fails.

Reading path: workload and capacity, pages 2–6; operations, recovery, and commercial terms, pages 7–11; fictional planning decision, pages 12–13; decision brief and next conversation, pages 14–15; sources, page 16. PDF page references may differ in Word.

02 / DEFINE THE WORKLOAD

Measure the work before sizing the system.

A useful workload profile describes ordinary demand, difficult cases, and peaks. It does not rely on one impressive demonstration.

Workload element	Planning detail to record
Task and deliverable	The specific request type, expected output, reviewer, and excluded actions such as sending or updating records.
Users and concurrency	Named user groups, likely simultaneous users, administrative users, and whether work is interactive or queued.
Sources and size	Document types, typical and largest file sizes, page counts, source locations, and how often sources change.
Volume and timing	Requests per day or month, busy periods, acceptable queueing, and work that cannot wait for batch processing.
Processing pattern	Retrieval, comparison, summarization, or other operations, including repeated runs and human correction cycles.
Growth and change	Expected changes in users, document collections, or task types, with a date for reviewing the assumptions.

Use ranges and name the source of each estimate

If the bank has not measured the workflow, label the figures as planning estimates and state who supplied them. Separate observed history from forecast demand. A peak caused by an examination, quarter-end, or staffing shortage may matter more than an annual average.

Also identify inputs that make one request much harder than another: very large files, poor scans, many versions, or conflicting sources. Capacity based only on short, clean examples can fail when the real workload arrives. Keep those difficult cases in the test set.

Do not hide uncertainty in a precise-looking number.

“Approximately 20–40 procedure questions on a routine day, with occasional peaks above 80, based on three months of ticket history” is more useful than an unsupported claim of exactly 31.7 daily requests.

This guide does not prescribe a universal request volume, document size, or hardware configuration. The bank and its technical reviewers determine what is material for the intended service.

03 / CHOOSE THE OPERATING BOUNDARY

Name every path the deployment can use.

Bank-hosted processing, optional online research, application connections, and media services are different operating arrangements. Include only the paths the bank intends to evaluate.

Path	What it may involve	Planning implication
Local processing	Approved files processed in the bank-controlled environment.	Test the actual configuration, capacity, records, and recovery.
Online research	Queries or content sent to an external information service.	Approve the information that may leave and the provider's terms.
Application connection	Reading or changing data through another system.	Separate viewing from updates and test permissions and effects.
Media services	Image, audio, video, or other specialized processing.	Confirm whether it is included, where it runs, and its data handling.
Administration and support	Updates, monitoring, diagnostics, backups, and remote assistance.	Define who can connect, what they can see, and how access ends.

Record paths as included, unavailable, or not yet evaluated

An optional feature can remain installed while being unavailable to the approved workload. Test the deployed restriction rather than relying on a menu label or an instruction in a prompt. If a path has not been assessed, it is not approved for sensitive information merely because another path was.

The deployment description should also identify source systems, identity providers, file storage, logging, monitoring, backup, and network dependencies. A service can fail because one of those dependencies fails even when the AI component itself is available.

Start with the smallest useful boundary.

For a first evaluation, a controlled local document workflow may be enough. Keep external research, application updates, customer communications, and specialized media unavailable until each has its own evidence and approval.

This is a planning distinction, not a description of every Providensi configuration. Included features, local execution, and external paths must be confirmed for the proposed deployment. [1, 2]

04 / TEST CAPACITY HONESTLY

A server specification is a hypothesis.

Proposed hardware, model size, and vendor estimates can guide a test. They do not prove that the bank’s workload will meet its service expectations.

Build a representative test

Use the proposed configuration, representative files, expected user activity, and the same major processing steps as the intended service. Include ordinary requests, difficult files, concurrent use, and repeated runs. Synthetic content can protect information during early testing, but it should resemble the real workload in size and complexity.

Capacity evidence	What to record
Configuration	Hardware, software release, model or service components, storage, network, and relevant settings. Record changes made during testing.
Workload	Number and type of requests, file sizes, concurrency, duration, and how the test differs from production expectations.
Service result	Elapsed time, queue behavior, completed and failed tasks, timeouts, and whether outputs met the functional checks.
Resource behavior	Processor, memory, storage, and other relevant utilization, including sustained peaks rather than only short spikes.
Operating impact	Effects on shared systems, administrator effort, logs, backups, and the ability to stop or recover a run.
Limitations	What was not tested, such as larger files, more users, external paths, or a failover environment.

Decide what the result supports

A test can support a limited workload while showing that a larger one needs more capacity, queue controls, or a narrower scope. Compare results with the bank’s service expectations and risk tolerance. If an objective cannot be met, revisit either the objective or the design rather than assuming later tuning will solve it. [3]

Capacity is not only speed.

Also test whether the environment can store the working files, retain required records, accept updates, and recover without corrupting accepted work. A fast answer is not useful if the operational path around it fails.

No hardware minimum or performance promise is stated here. NIST’s voluntary AI framework calls for evaluation in conditions resembling intended use, with documented limitations. [4]

05 / DESIGN THE RUNWAY FOR CHANGE

Leave room for the second workload.

A deployment sized only for today’s demonstration can become difficult to operate when users, documents, or retention needs grow.

Estimate a planning horizon with the sponsor, such as the pilot period and the first year of the intended service. Identify which changes are expected and which are only possible. New user groups, larger files, longer retention, more concurrency, and additional workflows should not all be treated as automatically supported by the initial design.

Planning choice	Question	Evidence or decision
Headroom	What increase can the tested design absorb before service quality becomes unacceptable?	Measured limit, monitoring threshold, and owner.
Expansion path	Can capacity be added, or does growth require a redesign or a reduced workload?	Documented technical options, lead time, and cost.
Shared resources	Will logs, backups, storage, or another application consume the assumed capacity?	Dependency review and capacity owner.
Change control	Who approves a new workflow, model, integration, or configuration change?	Impact assessment and retest criteria.
Review point	When will actual usage be compared with the estimate?	Named date, measures, and escalation route.

Use thresholds to trigger a decision

Thresholds should reflect the bank’s service needs, not a universal percentage copied from another organization. For example, the bank might review the design when sustained utilization, queue time, failed tasks, or storage growth reaches an agreed level. A threshold prompts investigation; it does not by itself prove that the service is unsafe or that more hardware is the right response.

Growth needs an owner and a lead time.

If adding capacity takes weeks, identify that constraint before promising a rapid expansion. Record whether the bank can temporarily limit users or workload while the change is made.

FFIEC infrastructure guidance treats architecture, infrastructure, and operations as related functions that support technology planning and risk management. It does not impose a numeric capacity rule. [2]

06 / CONNECT IDENTITY, SOURCES, AND OPERATIONS

Make the deployment operable by named people.

A technically successful installation still needs identities, source access, day-to-day administration, and a manual route when the service cannot be used.

Operating area	What to define before go-live
Identities	Who receives access, how accounts are issued and removed, which privileged roles exist, and how administrative actions are attributable.
Source access	Approved repositories, entitlements, access changes, and tests showing that denied users cannot retrieve restricted content through the service.
Work management	Where requests arrive, who operates the service, who reviews results, and how status is communicated.
Records	What logs, prompts, outputs, and decisions are retained; where they are stored; and who can read them.
Monitoring	Who watches availability, errors, capacity, access anomalies, and failed jobs, and what response time is expected.
Manual alternative	How staff complete urgent work when the service, a dependency, or the reviewer is unavailable.

Prove the difficult transitions

Test a new user, a departed user, a permission change, and an unavailable reviewer. Confirm that an administrator can stop a job and determine what completed. Check that source updates flow to the approved collection without bypassing ownership or version control.

Operational responsibility may be shared, but each recurring task needs a primary owner and a backup. A vendor can supply administration under an approved arrangement; that does not transfer accountability for the bank’s service or remove the need to understand what the administrator can access. [1, 5]

Do not let a shared login obscure responsibility.

Use accountable administrative access appropriate to the bank’s environment. Shared credentials make it harder to investigate changes, support activity, and unauthorized use.

These controls should fit the bank’s existing identity, operations, and records processes. The deployment plan coordinates them; it does not replace those approved procedures.

07 / SUPPORT WITHOUT OPENING EVERY DOOR

Support access is part of the deployment design.

The bank needs a workable way to diagnose problems. That need does not require standing, unrestricted access to customer information or production records.

Support decision	Practical boundary
Who may help	Name the bank team, provider, or both. Identify subcontractors who may participate.
How access starts	Require a request, an authorized approver, a stated purpose, and a defined duration.
What access includes	Separate viewing configuration from viewing content, exporting diagnostics, changing settings, or copying files.
How it is supervised	Use the approved channel, record activity, and give the bank a way to end the session.
What happens afterward	Revoke access, confirm it ended, and retain the record required by the bank’s process.

Design diagnostics before an incident

Agree on the minimum diagnostic information needed for common failures. Where possible, keep sensitive file contents out of routine logs and support bundles. If a difficult problem requires content access, treat that as a separate approval with a narrower scope and a shorter duration.

Test the support route before relying on it. A telephone number in a proposal is not evidence that the right people can respond, that the access method works, or that the bank can revoke it. Confirm hours, severity definitions, escalation contacts, and any dependence on a third-party portal or network path.

Match support terms to the role of the service.

A limited evaluation may accept next-business-day assistance. A service used for time-critical work may need a faster commitment, an on-site bank administrator, or a manual process. Do not infer the support level from the product name.

Support and maintenance are part of the system’s useful life, not a purchasing afterthought. FFIEC guidance emphasizes maintenance and resilience across that life cycle. [5]

08 / UPDATES, RELEASES, AND CHANGE

Every material change needs an owner and a retest decision.

A new model, software release, source collection, connector, or configuration setting can change quality, capacity, and the information boundary.

Identify who may propose a change, who assesses its impact, who approves it, and who can reverse it. Include bank-managed configuration as well as provider-supplied releases. A release schedule should tell the bank what is changing in time to assess it, without assuming that every update requires the same depth of review.

Change type	Questions before production use
Software or model update	What changed, which workloads are affected, and which quality, security, and capacity tests must be repeated?
Configuration change	Does it alter access, routing, retention, logging, or an optional external path?
Source or workflow change	Are new information types, users, or actions involved, and is the reviewer still appropriate?
Infrastructure change	Does it affect capacity, dependencies, backups, or the tested recovery path?
Emergency change	Who may authorize it, how is it recorded, and what review follows after service is stabilized?

Preserve a workable return path

Before applying a material release, know how to return to the last accepted version or continue through the manual process. Confirm that rollback will not damage records, source files, or queued work. A release that cannot be reversed safely needs a stronger pre-release test and an explicit decision to accept that constraint.

Retest the behavior that the change could affect.

A small interface update may need a limited check. A new integration or processing route can require access, quality, capacity, and recovery tests. Record why the selected test is sufficient.

Change management should use the bank’s existing technology governance. This guide adds AI-specific questions; it does not create a separate approval hierarchy or promise compatibility after every update. [4, 5]

09 / INTERRUPTIONS AND RECOVERY

Define the service you must restore.

Recovery planning starts with the business impact of losing the workflow, then examines data, dependencies, and a realistic fallback.

Ask what work stops, what can wait, and what must continue manually if the service is unavailable. The answer may differ for an evaluation system and a production service used during a deadline. Use that impact to set recovery objectives rather than adopting a generic infrastructure target.

Planning measure	How to use it
Business impact	Identify affected users, deadlines, customer or operational consequences, and manual alternatives.
Recovery time objective (RTO)	Define how quickly the specified resource or service must return to a usable state, within business-impact tolerances. Include dependencies and checks, not just server start-up.
Recovery point objective (RPO)	Define the required recovery point before disruption, commonly expressed as the maximum acceptable age of recoverable data. Identify which records it covers and how lost work is reconciled.
Dependencies	Include identity, storage, network, source systems, providers, facilities, and people required for restoration.
Test result	Record the scenario, what was restored, elapsed time, data loss, problems, and corrective actions.

Restore the workflow, not only the machine

A recovered server may still lack the approved source collection, current permissions, model components, or the last accepted outputs. Define the minimum restored state that lets staff work safely. Verify backups and practice the relevant recovery path before relying on the plan. [3]

For an interrupted AI job, determine what finished, what did not, and whether repeating it could duplicate an action. In a preparation-only workflow, preserve the draft state and have a reviewer inspect it. If a future workflow can send or update records, reconcile those effects before retrying.

A fallback is part of resilience.

If the bank can safely pause the AI workflow and use an existing manual process, document who makes that decision and how pending work is tracked. Not every internal assistance service requires immediate automated failover.

The FFIEC continuity booklet defines RTO and RPO and explains their use in business-impact analysis. It describes practices and does not impose universal recovery times. [3]

10 / LICENSING, EXIT, AND OWNERSHIP

Know what the bank can keep and how it can leave.

A deployment decision should identify the rights, restrictions, and practical steps required to change or end the arrangement.

Decision area	What reviewers should make explicit
License and use	Who may use the software or service, in which environment, and whether affiliates, vendors, or standby systems are included.
Model and components	Which components the bank receives, hosts, or accesses, and which use restrictions or third-party terms apply.
Bank content	Rights and restrictions covering inputs, prompts, outputs, fine-tuning material, logs, and files created by staff.
Portability	Whether accepted work, configuration, and records can be exported in a usable form without the original service.
Exit assistance	Access revocation, deletion or return obligations, transition period, and support available during the move.
Unresolved restrictions	Any term that could prevent the intended use, audit, recovery, or replacement, with its owner and decision.

Test the practical exit, not just the clause

Export a sample of accepted work and required records, then confirm that another authorized environment or a manual process can use them. Identify formats, missing metadata, and dependencies that would make the export incomplete. A contractual right is less useful if the bank cannot produce a readable copy when it needs one.

Also plan the operational side: disabling accounts, ending provider access, retaining what the bank must retain, and handling copies held by subprocessors. Legal and sourcing professionals should review the actual agreement. This planning guide cannot determine whether a clause is acceptable for a particular bank. [1]

Put ownership questions in the proposal before signature.

Ask which environments, standby copies, support parties, and future workflows are covered. Changing those assumptions after installation can create both a technical project and a contract amendment.

No ownership of a model or unrestricted right to use bank content should be assumed from a product category or a sales description. The controlling terms and technical evidence decide those questions.

11 / FICTIONAL PLAN: SCOPE AND CAPACITY

Test the proposed workload.

Every quantity, test result, and decision in this example is invented. It is not a bank case study, hardware recommendation, or observed Providensi performance.

A fictional bank plans a procedure-comparison evaluation for six requesters, two operators, a policy reviewer, and a backup. Invented ticket history suggests 20–40 daily requests with peaks above 80. Those are demand assumptions, not measured AI throughput. Early work uses only approved synthetic procedures in a bank-controlled test environment.

Test element	Fictional evidence	What the finding supports
Representative set	100 comparisons; at most two text PDFs per job, each 1–20 MB and at most 80 pages. Five jobs run concurrently.	Only this task, file range, and recorded configuration. Scans and larger files remain untested.
Timing criterion	At least 95 of 100 jobs must produce a draft within 8 minutes of queue entry. In the example, 97 do.	97% meets the invented timing criterion. It is not a guarantee of human-reviewed completion.
Quality criterion	At least 95 of 100 drafts must pass the defined source-and-completeness review first time, with no critical error observed. In the example, 96 pass; four are corrected.	96% meets the invented quality criterion. No critical errors observed does not prove none can occur.
Higher concurrency	A separate 100-job run with the same file profile and ten concurrent jobs has only 83 drafts ready within 8 minutes.	83% misses the criterion. Ten-job concurrency is not approved.
Resource and queue record	No resource-exhaustion failure occurs in the five-job test; CPU, memory, storage, and queue records are retained.	Useful evidence for this test, not proof of untested growth headroom.

The capacity decision

The team keeps a five-active-job limit and routes files outside the tested profile to the manual process. A heavy queue triggers operator review instead of an unconditional eight-minute promise. The failed ten-job test becomes an open scaling item with the technology lead, not a reason to hide the result or assume more hardware is already justified.

Draft speed and service capacity are different.

The timing clock includes queueing and processing but ends before human review. Reviewer availability and total staff effort still determine how quickly acceptable work reaches the requester. The daily demand estimate has not been proved sustainable by this batch test.

Every threshold here was invented for teaching. The bank chooses its own criteria before testing and records sample limits, uncertainty, and the consequences of error. [4]

12 / FICTIONAL PLAN: THE DECISION

Approve the next stage, not every future use.

The invented example now adds operating evidence and an explicit evaluation decision. It does not authorize a real deployment or supply generally appropriate recovery targets.

Fictional evidence	Recorded finding
Boundaries and identities	Tests deny external research, source writes, system-record changes, and assistant sends. A removed user and an unauthorized identity cannot retrieve the synthetic source or derived answer.
Records and terms	Trial logs, indexes/caches, backups, licenses, and permitted data use are documented for the synthetic evaluation. Production handling remains a separate review.
Recovery exercise	The chosen trial RTO is four hours; RPO permits a recovery point no older than four hours before outage. A bank administrator restores the tested usable state in 2 hours 45 minutes, from a two-hour-old recovery point.
Support and fallback	Bank staff restore without provider help. Next-business-day provider support is not the recovery promise. A separately authorized synthetic support session is logged and revoked; no standing access remains.
Exit and review	Sample accepted work and source references open outside the service. The policy reviewer and backup have access and scheduled review capacity; manual handling remains available.

The recorded evaluation decision

The authorized bank group permits a 30-day synthetic-only evaluation for the named team and tested file profile, with at most five active jobs. Real bank documents, customer information, external research, record changes, and assistant sends remain excluded. Temporary provider access still needs a separate approval for each session. Nothing is approved for production.

The service owner reviews the queue, acceptance, correction effort, and coverage weekly. Stop affected work on unapproved access, transfer, action, or configuration change. Route excess load manually and investigate recurring timing or quality misses. The technology lead owns the unresolved scaling issue. Authority expires after 30 days unless the designated group renews it after review.

A recovery point is not a promise of no lost work.

The fictional exercise leaves two hours of synthetic work outside the restored copy. Staff identify affected requests, mark uncertain drafts, and reconstruct them through the manual route before relying on them. Recovery of the tested state is distinct from clearing that backlog.

The four-hour objectives and all outcomes are teaching assumptions for this bounded service. One exercise does not prove recovery for every incident, outside the tested coverage, or for a more critical workload. [3]

13 / SUPPORTING TOOL: DEPLOYMENT DECISION BRIEF

Bring the decision onto one page.

Complete this brief after the planning discussion. Use it to expose assumptions and evidence before a proposal, pilot, or production decision is finalized.

Decision item	Entry or evidence reference
Workflow, users, and excluded actions	
Volume, file range, and source of estimate	
Environment and included or unavailable paths	
Capacity result and untested conditions	
Identities, source access, and operations owners	
Support route, duration, and revocation evidence	
RTO, RPO, fallback, and latest test result	
Licensing, data rights, portability, and exit gaps	
Decision, scope, review date, and approver	
Conditions that require reassessment	

Before relying on the brief

- Each material claim has an evidence reference, a date, and an owner.
- Estimates, test results, and production approvals are clearly distinguished.
- Unavailable or untested paths are not described as approved.
- The manual route and recovery responsibilities are specific enough to use.

This brief is a discussion and decision aid. It does not authorize production use, create a contract, or prove that a control operated. The Word edition is editable; the PDF is intended for printing or annotation.

14 / USE THE PLAN IN THE NEXT CONVERSATION

Ask for a proposal that matches the workload.

A useful vendor conversation starts with the bank's operating requirements and ends with the evidence still needed before a decision.

Bring the intended workflow, user group, document profile, expected volume, information restrictions, support expectations, and recovery needs. Ask which of those requirements the proposed bank-hosted arrangement includes and which require separate configuration, licensing, infrastructure, or services.

Questions worth settling early

- Which components run in the bank's environment, and which optional paths remain available?
- What representative workload will be used to evaluate capacity, and who supplies the acceptance criteria?
- How are identities, source permissions, logging, backups, and administrative access handled?
- What support and release commitments apply to the proposed service level?
- Which records and work products can the bank export if it ends the arrangement?

Ask for the whole cost picture

Separate initial infrastructure, configuration, migration, testing, and staff training from recurring licenses, usage, monitoring, support, backup, and review effort. Identify replacement, upgrade, incident-assistance, and exit costs where relevant. Label estimates and contractual prices separately. A proposal that prices only the model or server does not establish the full operating cost.

A deployment discussion is not a deployment approval.

Use the answers to identify owners, tests, and contract questions. The bank's technology, security, sourcing, and business decision makers determine whether the evidence is sufficient for the next stage.

This planning guide complements the earlier resources on practical AI use, value measurement, supervised internal operations, and due diligence. Those guides help define the work and evidence; this one connects them to the operating environment. None of the resources establishes that a particular deployment is appropriate for every bank.

Bring synthetic or approved examples to an initial discussion. Do not send customer records, credentials, or confidential production files merely to request a planning conversation.

SOURCES / SCOPE

References and reading notes.

References checked October 6, 2026. The planning method, fictional example, and decision brief are Providensi's educational synthesis, not examiner procedures or vendor requirements.

[1] Third-Party Risk Management: A Guide for Community Banks

Federal Reserve, FDIC, and OCC • May 2024

A resource for planning, due diligence, contracting, monitoring, and ending third-party relationships. It reinforces that bank responsibilities continue and does not create a checklist or safe harbor.

[2] Architecture, Infrastructure, and Operations

FFIEC booklet; OCC Bulletin 2021-30 • June 30, 2021

The OCC notice describes examiner guidance on governance and risk management for technology architecture, infrastructure, and operations, including connected environments and third-party dependencies. No hardware configuration is prescribed by the cited notice.

[3] Business Continuity Management

FFIEC IT Examination Handbook • November 2019

Describes business-impact analysis, recovery objectives, dependencies, resilience, and testing practices. It expressly does not impose requirements or universal recovery times. See printed pages 9–12, corresponding to PDF pages 12–15, for impact analysis and RTO/RPO.

[4] AI Risk Management Framework (AI RMF 1.0)

NIST • January 26, 2023

A voluntary framework. Its Map, Measure, and Manage functions support evaluation in the intended context, documented limitations, monitoring, and an informed decision about whether to proceed.

[5] Development, Acquisition, and Maintenance

FFIEC booklet; OCC Bulletin 2024-26 • September 5, 2024

The dated OCC notice describes examiner guidance on development, acquisition, and maintenance across a system's useful life, including project management, supply-chain considerations, and resilience. The notice date is not represented as the booklet's original publication date.

Educational scope

This guide is not legal, sourcing, security, or engineering advice and does not prove that any configuration is suitable or compliant. The bank's responsible teams must assess its workload, obligations, risk appetite, contracts, and technical evidence. No Providensi feature, hardware minimum, support commitment, or recovery outcome should be inferred from these examples.